



DIGITAL SAT CALCULATOR GUIDE

GUIDE TO USING THE BUILT-IN CALCULATOR

Section 1	Scientific Calculator
Section 2	Graphing Calculator
Section 3	Scientific Calculator Example Questions
Section 4	Graphing Calculator Example Questions

INTRODUCING THE BUILT-IN CALCULATOR

The Desmos calculator that is built into the Bluebook testing app can be enormously helpful on the Digital SAT. The more you become familiar with it, the faster you'll be with it. You will also have a better understanding of when it can help a little, when it can help a lot, and when it won't help at all.

One Tool Among Many

When you use a calculator, try not to rely on it too much. It can't interpret a Digital SAT Math question for you, so continue to read carefully. It also doesn't have AI (yet), so it does exactly what you tell it to do. Write things down on your scratch paper and double-check that you entered everything correctly.

How to Use This Guide

You may choose to work through this document all at once to become an expert at using the built-in calculator. You can also read it in pieces and come back anytime you want a refresher. We recommend that you read the Scientific Calculator section before working through the Fun with Fundamentals chapter in your *Digital SAT Prep* book, and the Graphing Calculator section fits best before the Functions and Graphs chapter in your *Digital SAT Prep* book.

Example Questions

Sections 3 and 4 contain example questions to show you in detail how the calculator can help. Read Section 3 after completing Section 1, and read Section 4 after completing Section 2. These questions come from Test 1 in your *Digital SAT Prep* book, which should have been the first thing you did. If you have not taken Test 1 yet, stop! Take that test first to get a realistic baseline of your strengths and weaknesses (and to avoid spoilers for the explanations in this section), then come back and see what kind of difference the calculator can make.

Can I Use My Own Calculator?

You certainly can! The important thing is to get a lot of practice with the calculator you intend to use on test day. If you're going to bring your own calculator, make sure it has fresh batteries and check that it's on the approved list on College Board's website. Even if you plan to use your own calculator, this document might give you some new ideas about how to use it to work questions on the Digital SAT.

Section 1

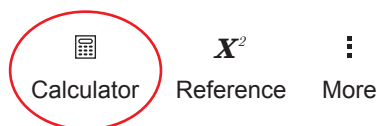
The Scientific Calculator

This section will introduce you to the basics so you can access the built-in calculator quickly and use it to solve math questions on the Digital SAT. A separate section will show you how to take full advantage of the graphing calculator. Take a moment to open the Test Preview section of the Bluebook app and get to a math question so you can try things out along the way.

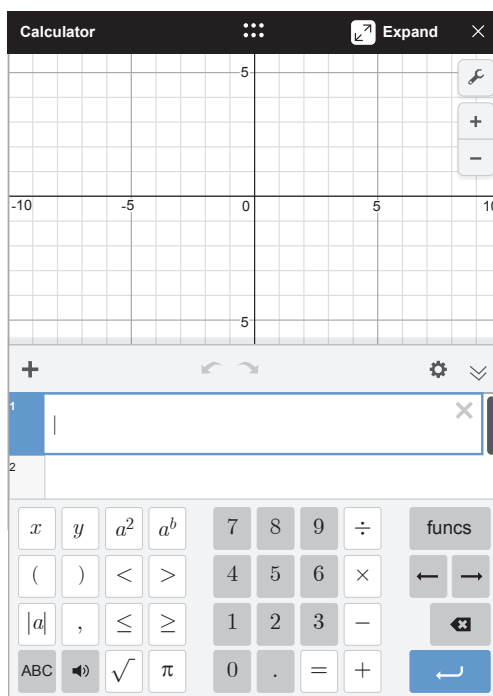
THE CALCULATOR WINDOW

How to Open the Calculator

To open the calculator, click the “calculator” icon in the upper right corner of the screen.



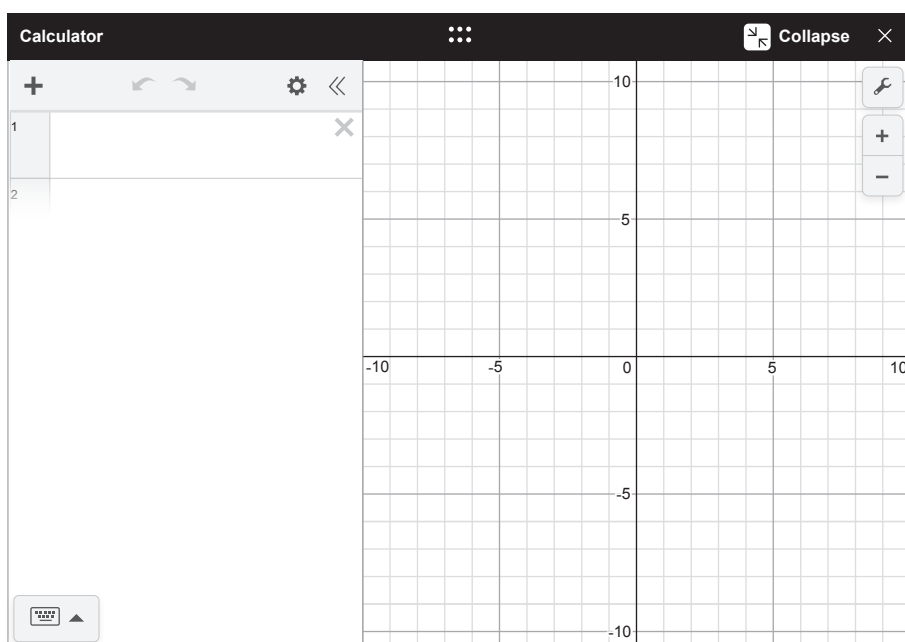
The calculator will open set to a default compact size, showing a small section of the graph display, an entry field, and a 123 keypad with entry buttons.



The default compact size appears on the left side of the screen and takes up about one-fourth of the screen area. In this format, the graph display is on top, the entry field is in the middle, and the entry buttons are below.

How to Expand the Calculator

To expand the calculator panel, click the “Expand” button located in the upper right corner. This will expand the panel so that the graph display is located to the right and the entry field is located to the left. In this view, the “Expand” button has been replaced with a “Collapse” button. The expanded calculator takes up about three-fifths of the screen.



To return the calculator to the default compact size, click the “Collapse” button in the upper right corner.

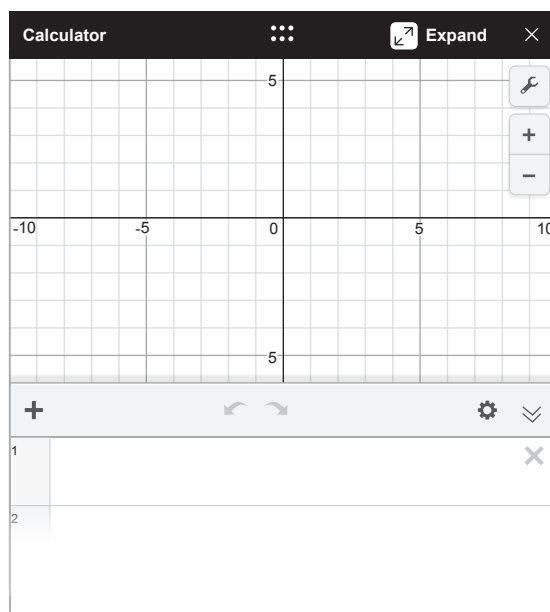
To close the calculator completely, click the X in the upper right corner. Note that closing the calculator will clear everything that’s been entered.

How to Access the Keypad in Expanded Mode

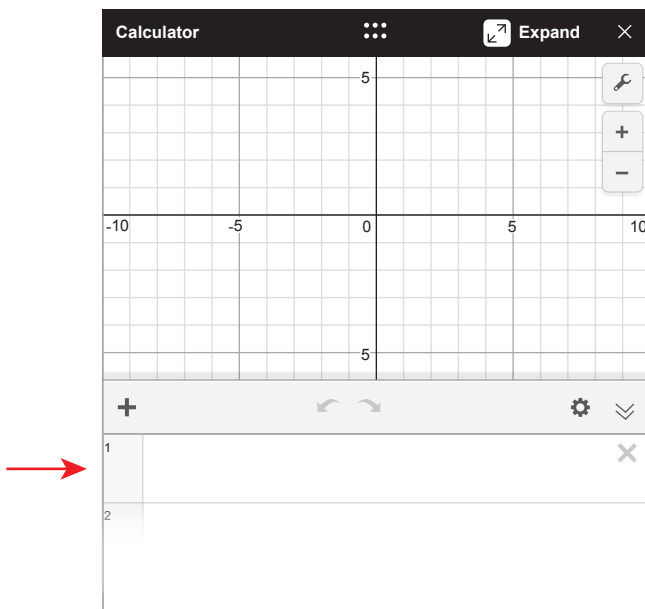
Notice that the keypad with the entry buttons will disappear or be minimized when the calculator is expanded. To bring up the 123 keypad again, click the small icon of a keyboard at the bottom left corner of the panel:



The entry buttons may also disappear when the calculator is collapsed after being expanded or when the calculator is dragged to a new position. In this case, there will not be a keyboard icon:

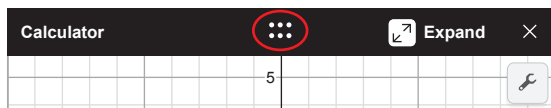


To make the entry buttons visible again, click in an entry field:



How to Move the Calculator

To move the calculator, hover the mouse pointer above the dots on the top black border.



At that point, the regular mouse pointer will change into a “move” mouse pointer with arrows pointing in four directions.

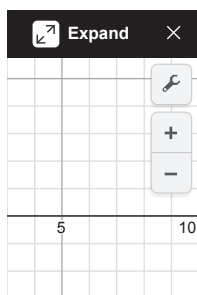


Click and hold on the six dots. The “move” mouse pointer will change into a “hand” mouse pointer.



Then, while continuing to hold, drag the calculator into a new position.

It is possible to drag the calculator so that a part of it is off the screen, possibly obscuring the “keyboard” icon that displays the entry buttons or the Expand/Collapse, Graph Settings, Zoom In, and Zoom Out buttons, which are located on the far upper right of the calculator.



If any needed buttons are not visible, try moving the calculator as described above until all edges are visible on the screen. The keyboard shortcuts below can also be used to open or close the calculator:

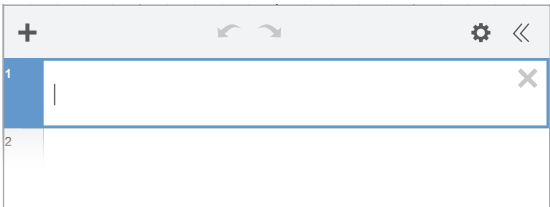
PC or Chromebook: Ctrl + Alt + C

Mac or iPad: Command + Option + C

ENTERING MATH

The Entry Field Toolbar

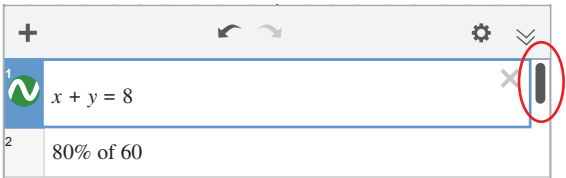
The expression list is the area where you enter calculations. It starts with two entry fields and adds more as you use them. There is a toolbar across the top:



Here’s what each button on the toolbar is for:

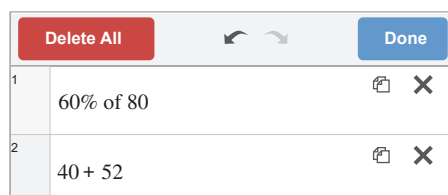
Button	Function
	Add item to field
	Undo last action
	Redo last undone action
or	Hide expression list
	Clear this entry field
	Edit the field list

Each entry field has a scroll bar to allow different calculations to be processed simultaneously.



If you use more than one entry field in the expression list, it will be easier to see in Expanded view.

Clicking the “edit field list” button brings up additional options: delete all fields, duplicate a single field, or delete a single field.



The entry field will display any alphanumeric characters entered into it, as well as math operators. When you enter a calculation in the entry field, the result is displayed in the lower right-hand corner of that entry field. For an ongoing or large calculation, the result will update as you enter each new term or operator. If the result of a calculation is too large to be displayed, the entry field will show the result as “undefined.”

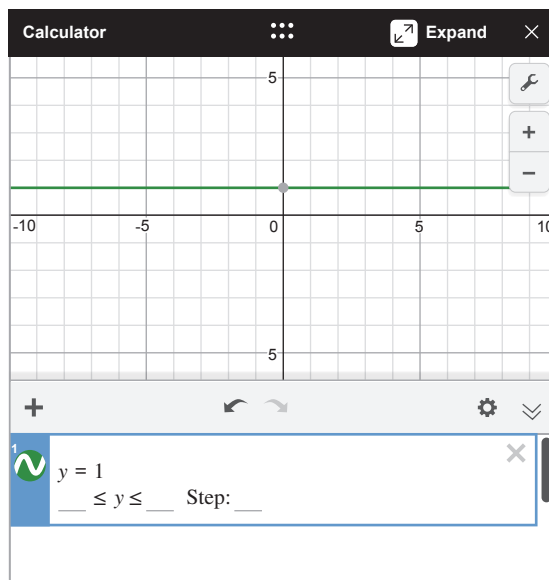


When you click the  button, you will have the option of entering an expression or a table, but the default is an expression. Some questions are about tables of values, so you might use that option on a few questions. However, you will mostly enter expressions, so let's focus on that.

Want to graph an expression or equation that uses variables other than x or y ? Just substitute x and/or y for the variables in the question.

Entering Expressions with Variables

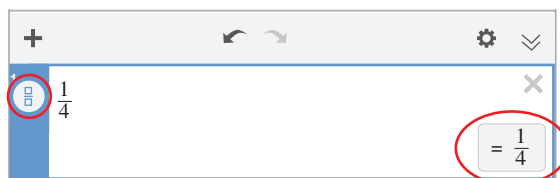
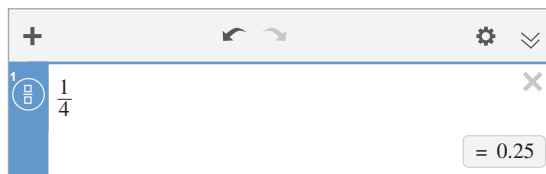
When a simple linear equation with either x or y is typed into the entry field, the graph display will show a graph of the expression or equation.



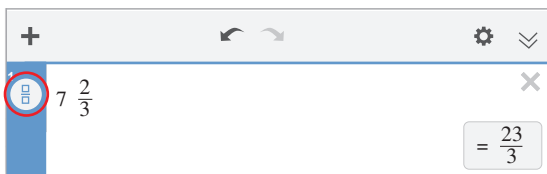
When entering a linear equation with any variable, the entry field displays a “range” option that allows the user to explore a range of values for the variable entered. However, only expressions or equations with x and y will generate a graph in the graph display.

Number Format

When you enter a calculation involving division, the entry field will represent the operation as a fraction and will display the result as a decimal by default. To convert the decimal into a fraction, click on the small symbol in the left blue border.



To enter a mixed number, enter the integer and fractional parts next to each other. To convert the mixed number into an improper fraction, click on the small symbol in the left blue border.

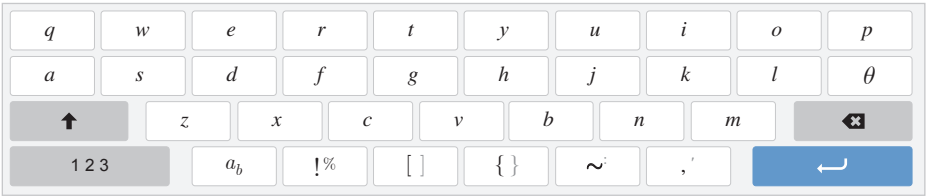


Built-in Calculator Keypad Options

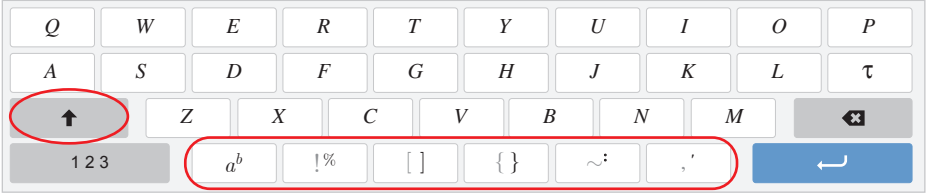
There are some basic navigation buttons on the keypad:

Button	Function
 or 	Bring up the functions menu
	Delete prior character in entered expression
 and 	Move cursor to the left or right in entry field
	Create new entry field
 or 	Switch between numeric and alphabetic keyboards

Clicking the “ABC” button toggles the keypad to the ABC keypad. The percent (%) button is located on this keypad, as well as the theta symbol and the 26 letters of the Latin alphabet, plus more bracket options:



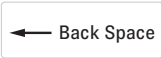
There is also a Shift button option, which will make the letters upper case and will toggle between subscript to superscript, factorial to percent, and open to closed brackets.



USING THE COMPUTER KEYBOARD

Physical Keyboard vs. On-screen Keypad

It is possible to enter a calculation into the entry field by typing on your physical computer keyboard and is very likely faster than clicking buttons on the on-screen keypad with your mouse or trackpad. Experiment with both methods and find the one that works best for you. Here are some computer keyboard keys that perform the same actions as their on-screen keypad button counterparts:

Built-in calculator keypad	Computer keyboard (MAC)	Computer keyboard (PC)
		
		
		
		

You can use a computer keyboard to enter numbers and letters, and many math operations can be entered using a computer keyboard. Here are some of the key keys to know:

Computer keyboard (MAC)	Computer keyboard (PC)	Function
		Addition
		Subtraction
		Multiplication
		Division
		Equals
		Exponent
		Absolute value
		Percent

Many of these functions require using the Shift key on your keyboard, and some of them will be in different places depending on the keyboard you're using. Be sure to practice with the device you'll be using on the test to avoid having to search for the right button during the real SAT!

KEYBOARD OPTIONS AND THEIR QUIRKS

Entering fractions

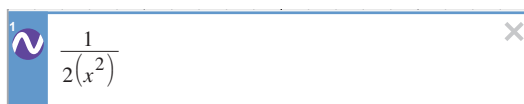
It's important to be careful when entering fractions into the entry field, especially when using a computer keyboard. Typing  for the division sign or fraction bar will cause the built-in calculator to treat every subsequent entry as part of the denominator.

Here's an example:

Say you want to enter “one-half of x squared” and you use the computer keyboard to type the following:

! 1 ? / @ 2 (9 x ^ 6 @ 2) 0

The result will be this incorrect entry in the entry field:



$$\frac{1}{2(x^2)}$$

Using the keyboard arrow keys can avoid this issue. To enter “one-half of x -squared” in the entry field correctly, type the following:

! 1 ? / @ 2 → (9 x ^ 6 @ 2 →) 0



$$\frac{1}{2}(x^2)$$

It is necessary to use another right arrow before the close parenthesis symbol or else the parenthesis will close around the exponent, which could cause problems.



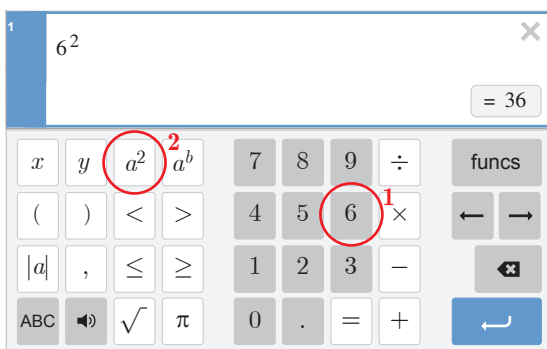
$$\frac{1}{2}(x^{(2)})$$

Notice that the entry field will suggest a closed parenthesis in a faint gray font if there isn't one entered. Typing the right arrow or clicking Enter will accept the suggestion and close the parentheses.

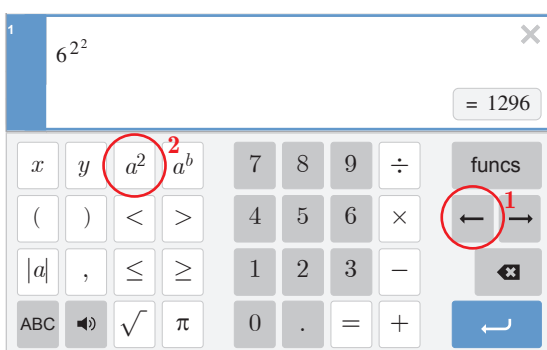
Another option is to use the 123 keypad on the built-in calculator to enter fractional expressions. However, you will still need to use the arrow buttons on the keypad to move the cursor out of the denominator if desired.

The “Square” Button

The primary use of the square button is to raise any number entered into the entry field to the second power, or to square it. To do this, type or click the number first, and then click the square button:

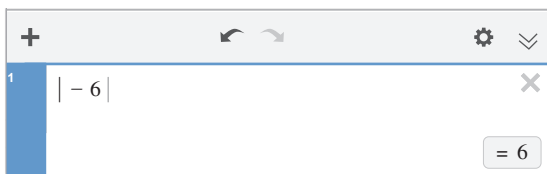


To square the result again, type or click the left arrow to put the cursor next to the exponent and then click the “square” button again.



Absolute Value

Entering only one | symbol in front of a number will return the absolute value of a number. If all that’s needed is the absolute value of a number, it’s not necessary to enter the closing | symbol.



However, if you want to enter additional math after the absolute value operation, you have to enter the closing $|$ symbol first. Otherwise, the built-in calculator will treat every quantity subsequently entered as subject to the absolute value sign. This is similar to how open versus closed parentheses behave in the calculator. Compare these results:

Calculator interface showing the expression $|-6| + 3$. The result is 9.

Calculator interface showing the expression $|-6 + 3|$. The result is 3.

Percent Defaults

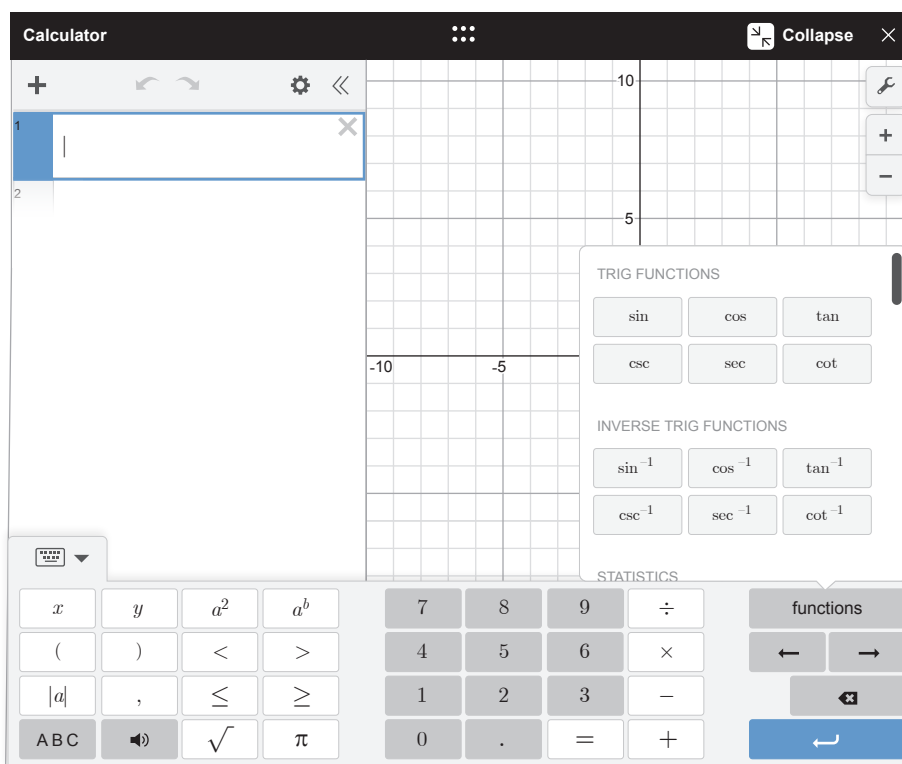
When you type the percent sign, the entry field will automatically add “of.” If you type the word “of” yourself, the result is a function, which is probably *not* what you wanted.

Calculator interface showing the expression 60% of of. The result is a function.

THE FUNCTIONS MENU

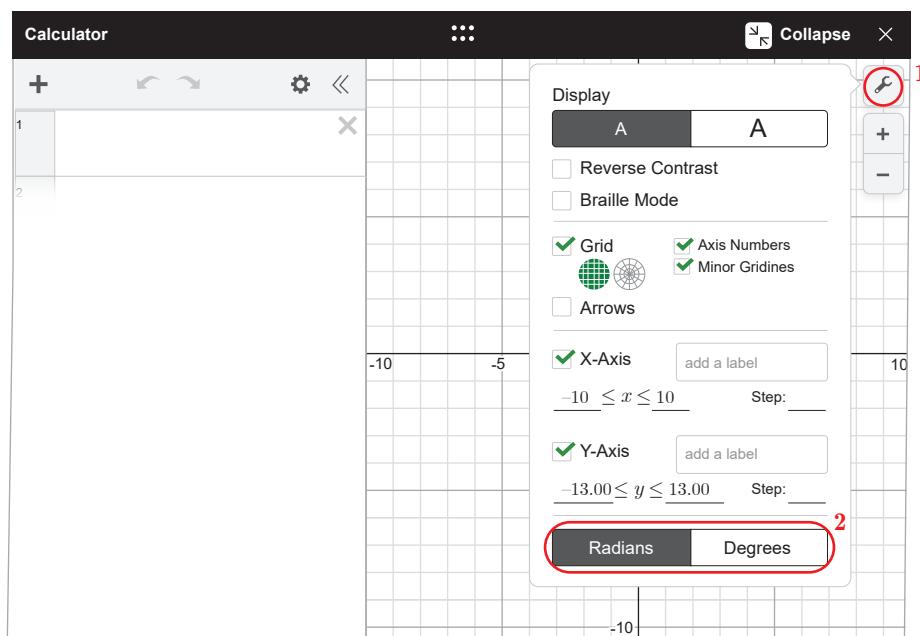
For the most part, the functions menu will not help on the Digital SAT. However, there are a couple of useful buttons, so let's take a look at those. To access the advanced functions, click on the **funcs** button when the calculator is its default size or the **functions** button when the calculator is expanded. Note that this button is *not* available on the ABC keypad, so go to the 123 keypad if you need to use the functions button.

The functions menu pops up with a scroll bar, but the menu will disappear if any other calculator area is clicked. Be aware of this since it is easy to visually confuse the functions menu scrollbar with the entry fields scrollbar.



Trigonometric Functions

The first set of buttons in the functions menu are for the trig functions. To use them, click on the button for the function first, and then enter the angle. Click on the wrench icon in the upper right in order to switch between radians (the default) and degrees.



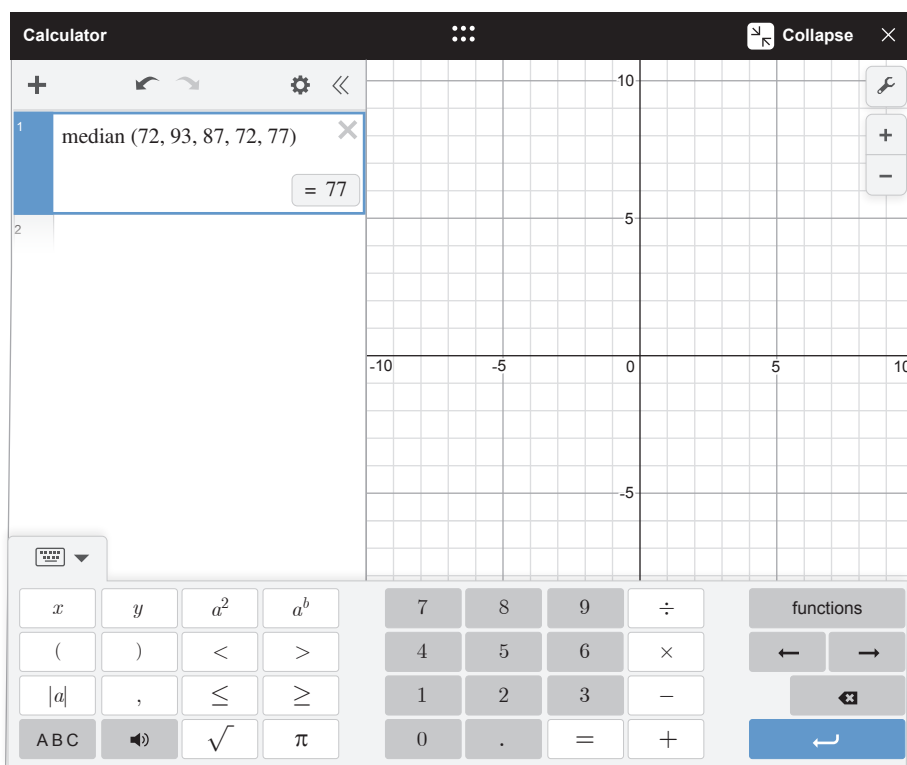
Trig functions can also be typed manually into the entry field.



While these buttons do exist and could be useful on one or two questions, be aware that most trig questions on the Digital SAT test the basics. If you know SOHCAHTOA and the conversion between radians and degrees, you probably won't need the calculator at all for such questions.

Statistics

Move the scroll bar in the functions menu to go a little lower, and you will see a set of buttons labeled “statistics.” The **mean** and **median** buttons will be useful. In both cases, click the button first, and then enter the list of numbers in the parentheses. Like trig functions, you can also type “mean” and “median” manually into the entry field to achieve the same result.



Conclusion

Now you know the basics of using the built-in calculator! Learn it; love it; practice using it during your preparation so you can achieve your best possible score on the Math section of the Digital SAT. Section 2 of this guide goes into detail about using the built-in calculator for questions about graphs, so come back to it when you start working with functions and coordinate geometry during your SAT journey.

Section 2

The Graphing Calculator

Graphing calculators can be incredibly helpful on the Digital SAT Math modules. This section will discuss using the built-in graphing calculator. You can also use your own graphing calculator on the test. If you do, be sure to read up on how to use all of its features. Furthermore, you may find that your calculator cannot perform some of the methods this section discusses. You'll want to spend some time getting comfortable with the features of the built-in calculator to make sure you're using the right tool for each task.

BUILT-IN CALCULATOR BASICS

This section will assume that you know the basics of the built-in calculator, including opening and resizing the calculator, entering equations with the on-screen keypads or your computer's physical keyboard, and working with fractions within the calculator. Section 1 in this guide covers those topics and more.

Zooming

One feature that will come in handy when using the calculator for graphing is the zoom tool. If your mouse has a mouse wheel, you can zoom in and out using the wheel. You can also pinch with two fingers on a trackpad, and these keys on the right side of the graph can also be used to zoom in and out:



You can click and hold in the graphing area and then move your mouse to reposition the graph. If you've moved the graph or changed the zoom, a home button will appear:

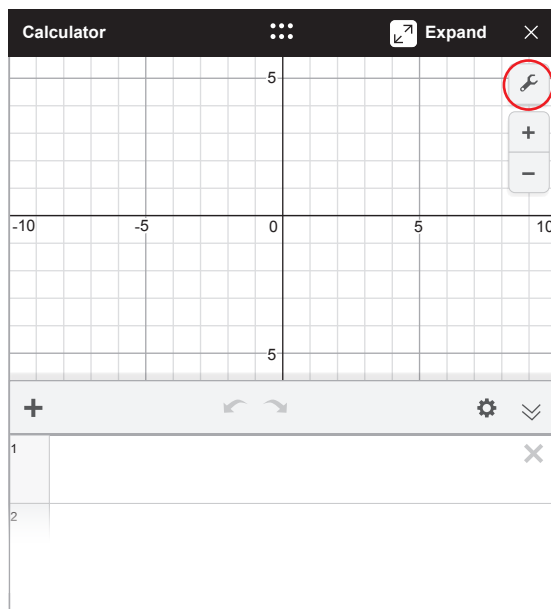


Clicking this button will reset the graph to the original view.

It is often easier to see features of the graph when the calculator is expanded. Experiment with the default view and expanded view, and use whichever makes things easier to see.

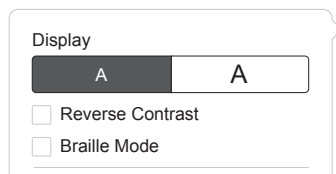
What's With the Wrench?

In the upper right corner of the built-in calculator, there is a wrench icon.

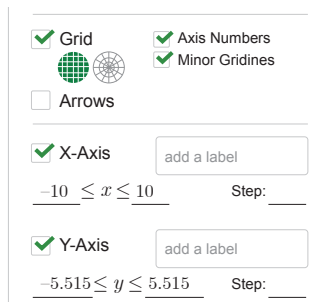


When you click on this icon, you will find a number of items that let you change the appearance of the graphing area.

The first set of options can help make the calculator easier to use. There is a larger font size available, dark mode (called “Reverse Contrast”), and Braille Mode. If you use a screen reader, take some time to ensure that all the settings you need are set up here, and practice using the built-in calculator with those settings.



The next set of options lets you change how the graph looks. You can turn the grid on and off. You can also change the grid into a radial grid; this isn't useful for the math that is tested on the Digital SAT, so we recommend keeping the grid as-is. We also recommend leaving the axis numbers and minor gridlines checked: doing so helps with Ballparking and otherwise identifying points and key features on the graph. Similarly, keep the x -axis and y -axis options checked. It is useful to use the ranges and steps to change how the graph looks, although you can zoom in and out using other tools to accomplish the same effect.



The screenshot shows the graph settings menu. It includes checkboxes for 'Grid' (checked), 'Arrows' (unchecked), 'Axis Numbers' (checked), and 'Minor Gridlines' (checked). Below these are sections for the 'X-Axis' and 'Y-Axis', each with a checked checkbox, a range input field, and a 'Step' input field. The X-Axis range is $-10 \leq x \leq 10$ and the Y-Axis range is $-5.515 \leq y \leq 5.515$. Each range field has an 'add a label' button next to it.

Adding a label just adds a label to the graph. It's a fun feature if you want to get fancy, but it's not really necessary for the Digital SAT.

The tool menu that you access by clicking the wrench icon is also where the built-in calculator hides the switch between Radians and Degrees.



The screenshot shows a unit selection menu with two buttons: 'Radians' (which is highlighted in dark grey) and 'Degrees' (which is white with a grey border).

The default is radians, so if you're working on a question that's dealing with angles in degrees, be sure to change the setting here. Furthermore, every time you close the calculator, it reverts to radians.

GENERATING GRAPHS

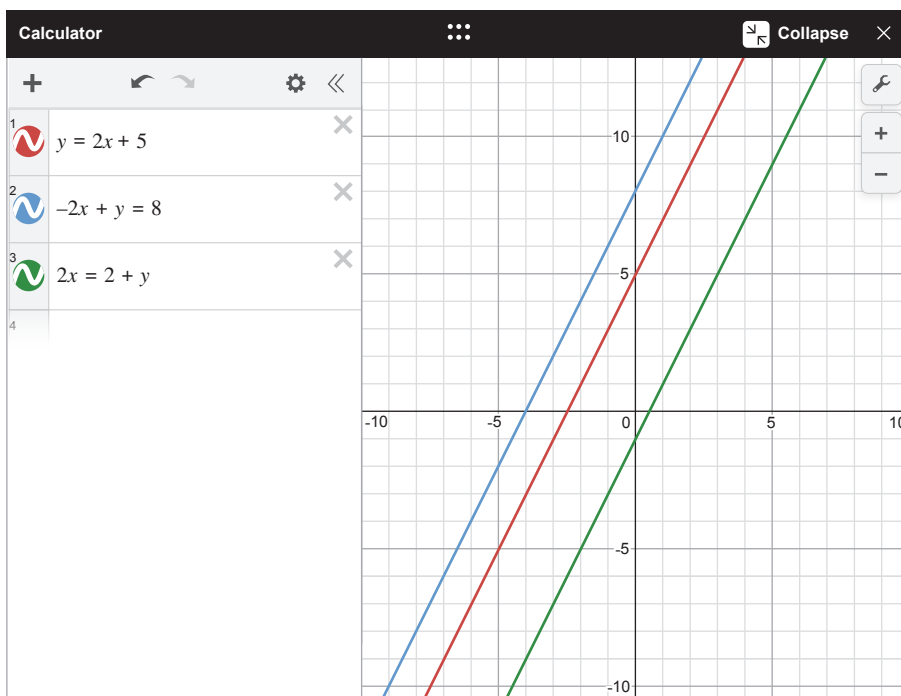
You saw in the Scientific Calculator section how to enter expressions and equations in the entry fields that make up the expression list. When the goal is to view a graph, there are a few additional things to keep in mind.

Why x and y ?

Equations that can be graphed express the relationship between variables. These will often, but not always, take the form of a function, in which $f(x) = y$. The Digital SAT may use letters other than x and y in equations like this. However, the built-in calculator will only graph an equation that uses x and y . If a question uses different letters, replace them with x and y . Write the equivalent letters on your scratch paper to make sure you enter everything correctly.

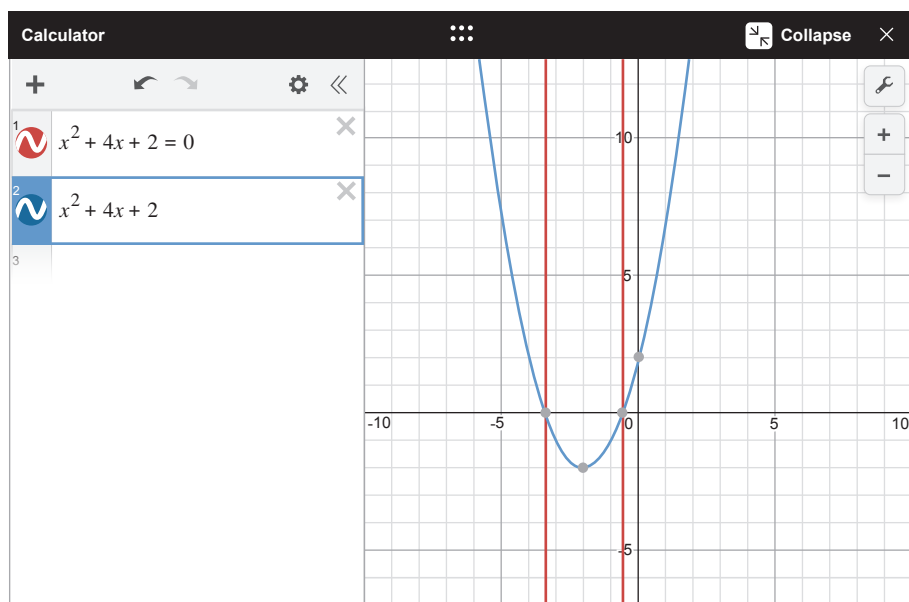
Form Isn't Always Function

One thing you *don't* have to worry about changing is the form of the equation or function. For example, the graphing calculator will show you a line if the equation is in slope-intercept form, standard form, or another form entirely. Take a look at the three parallel lines graphed below: the three equations look different, but the calculator graphed a line for each of them.



Graphing Quadratics

There is one thing that will change based on the form you enter. When graphing a quadratic, the result will look different depending on whether you set it equal to something (an equation) or don't (an expression). When solving a quadratic algebraically, you set it equal to 0 in order to find the values of x that make $y = 0$. When you enter a quadratic into the built-in calculator that way, the graph shows you two vertical lines. The points where those lines intercept the x -axis are the x -intercepts, or the solutions to the equation. That might look strange if you were expecting to see a parabola. If you leave out the “= 0” part of the equation, you will see a familiar parabola. Here are both versions graphed in the built-in calculator.



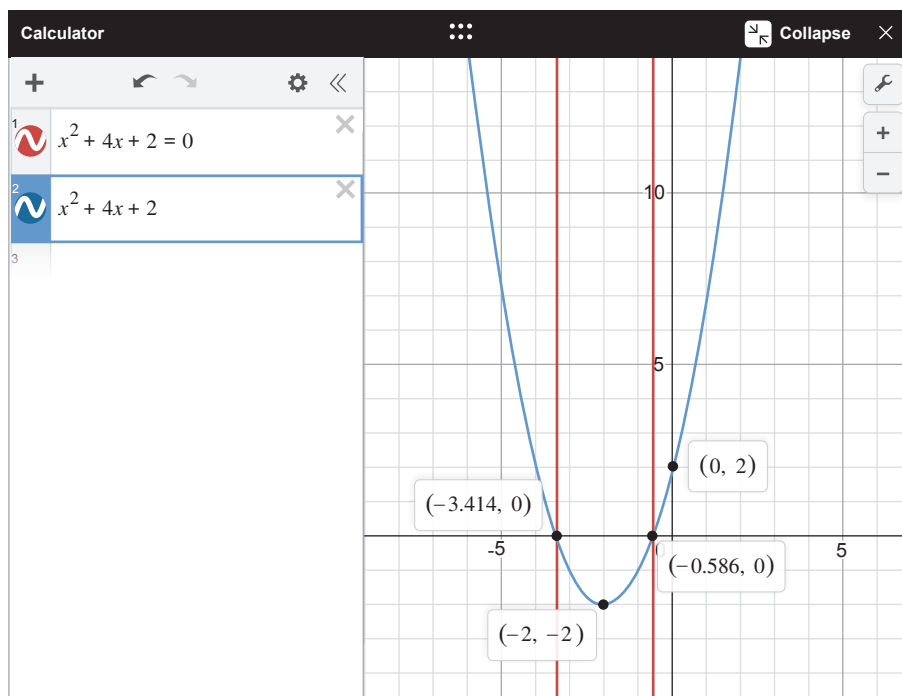
Notice that both graphs intercept the x -axis at the same two points, showing you the same information in different ways. If you prefer one appearance over the other, enter the quadratic that way. To generate a parabola, you might need to rearrange the equation to put everything on one side before entering it into the entry field.

POINTS ARE THE POINT

You might have noticed some gray dots in the previous example. These will help you find the answer to some Digital SAT questions quickly.

Where Is It?

Many questions on the Digital SAT ask about features of graphs in the xy -plane, most often the x - or y -intercepts, minimums, or maximums. That's what the gray dots represent. Clicking on a gray dot turns it black and adds the coordinates of the point next to it. Here's the previous example again so we can identify things.



The two dots along the x -axis are the x -intercepts. They are the same whether you graphed two vertical lines or a parabola. The dot at the bottom of the parabola is the minimum. The minimum is the vertex when the parabola opens upward. The maximum is the vertex when the parabola opens downward. The fourth dot on the graph above is the y -intercept, or the point where $x = 0$. This isn't tested as often, but it's good to know how to find it.

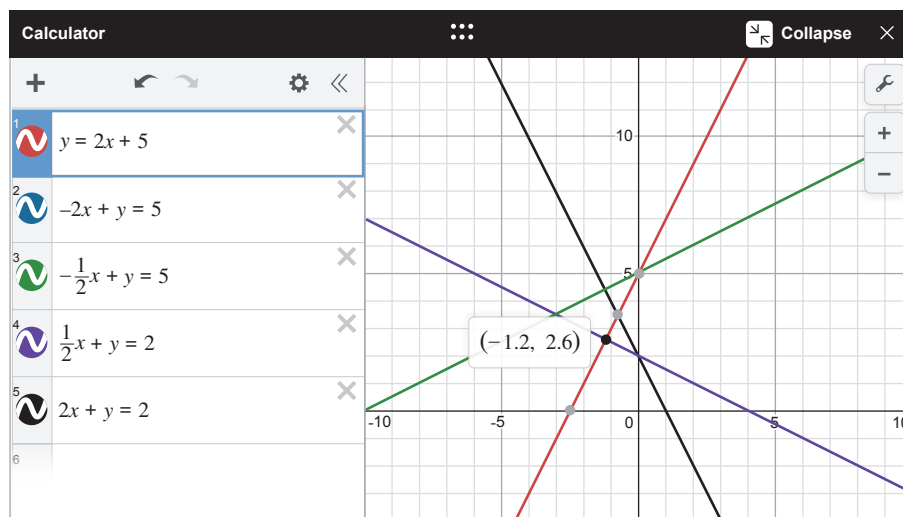
When Dots Disappear

The gray dots will disappear if you resize the graph or click somewhere else. To get them back, either click in one of the entry fields that has something in it or click on one of the colored lines or curves (not just in the graphing area).

More Equations? More Fun!

As you've seen, the graphing calculator will keep producing graphs as long as you keep entering equations or expressions that use x and y . Simply put each equation in its own entry field, and use the graphs to find features such as points of intersection. This is also a good way to evaluate the answer choices.

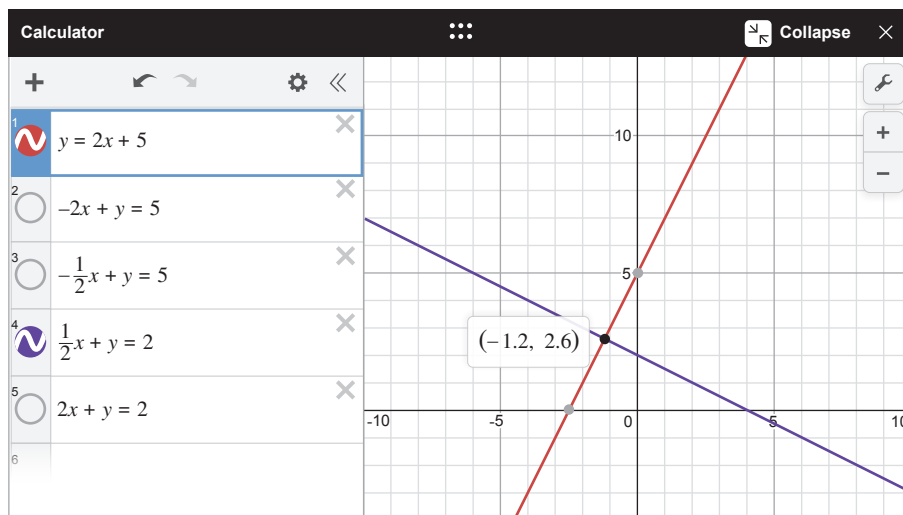
Let's say you're trying to find which answer choice has the equation of a line that is perpendicular to the line given by the equation $y = 2x + 5$. Simply enter all five equations and see what you have. This works even if the equations are in different forms.



The lines in entry fields 1 and 4 are perpendicular and intersect at $(-1.2, 2.6)$.

But wait, there are five equations but only four lines! That's because the first two equations are the same line in different forms. If you click in the first entry field and then the second, you will see one of the lines change color.

It's a good idea to enter the equation for the initial line—in this case, $y = 2x + 5$ —first, so you always know where it is. You can also click on the wavy symbol next to an equation to hide that graph, which makes it easier to see two at a time. Here's what the previous example looks like with only entry fields 1 and 4 selected.



Conclusion

There you have it! Those are the key features of the built-in graphing calculator. There are quite a few questions on the Digital SAT about functions and graphs, and a graphing calculator can be a game-changer. Keep in mind, however, that you still need to know the rules and be able to work things out by hand when necessary. If you know the underlying math *and* how to make productive use of the built-in calculator, you are well on the way to success.

Section 3

Scientific Calculator Example Questions

It's time to use the built-in calculator to do some Digital SAT Math questions! These questions come from your first practice test. If you used a calculator for these when you took the test, well done! Depending on which second module you took, some of these questions might be new to you. Read on to see how the built-in calculator can help answer questions quickly and efficiently.

Let's start with five questions that utilize some of the basic calculator functions. All of these come from the easier second module of Test 1.

1

Mark for Review

33, 34, 38, 41, 43, 44, 47

Which of the following is the median of the data listed above?

(A) 38

(B) 40

(C) 41

(D) 42

Here's How to Crack It

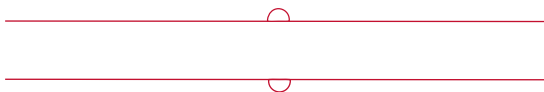
The question asks for the median of a set of data. The median of a list of numbers is the middle number when the numbers are arranged in order. In lists with an even number of numbers, the median is the average of the two middle numbers.

Either pull up the functions menu and click the **median** button or use the computer keyboard to type the word “median” followed by parentheses. Then, enter each number from the list inside the parentheses.

The computer will return the median of any list entered between parentheses after the word “median.”

A digital calculator interface with a blue header bar. The display shows the text "median(33,34,38,41,43,44,47)" in black. To the right of the display is a small "X" icon. Below the display, on the right side, is a button with the text "= 41".

That's it; you're done and got the question right! You didn't have to count, cross off numbers, or even make sure the list of numbers was in order. The calculator did a whole bunch of work for you. The correct answer is (C).



3



Mark for Review

A rectangle has a height of 23 inches (in) and a width of 9 in. What is its perimeter, in inches?

(A) 32

(B) 64

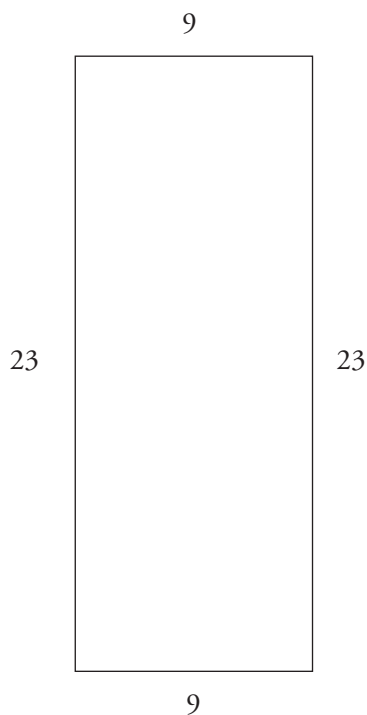
(C) 207

(D) 1,024

Here's How to Crack It

This is good example of how the calculator, the scratch paper, and your brain work together. The question asks for the perimeter of a rectangle, so Read The Final Question and make a note on your scratch paper that your goal is to find the perimeter. Then use the Geometry Basic Approach. Start by drawing a rectangle on the scratch paper. Next, label the figure with information from the question: length of 23 and width of 9. In a rectangle, opposite sides are equal, so this rectangle has two sides that are 23 inches long and two sides that are 9 inches long.

The drawing should look something like this:



Next, write down any formulas that will help. The formula for perimeter of a rectangle is $P = 2(l + w)$. Plug in 23 for the length and 9 for the width, and enter the formula in the calculator.

A digital calculator interface is shown. The input field contains the formula $2(23 + 9)$. To the right of the input field is a close button (X). Below the input field, the result $= 64$ is displayed.

You could also type in the value of each side one at a time to get $9 + 23 + 9 + 23 = 64$. Either way, the result is 64, and the correct answer is (B).



6  Mark for Review

For all positive values of y , the expression $\frac{3}{y+c}$ is equivalent to $\frac{15}{5y+30}$. What is the value of constant c ?

(A) 3

(B) 6

(C) 8

(D) 150

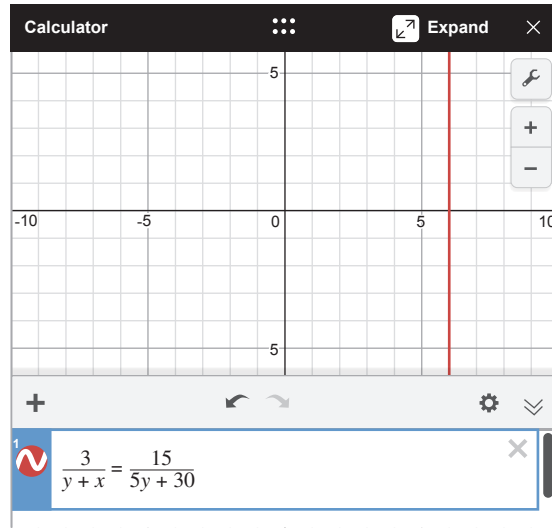
Here's How to Crack It

The question asks for the value of a constant given two equivalent expressions. This is an excellent opportunity to use the entry field to generate a visual solution in the graph display. However, the built-in calculator will only do this for the variables x and y , so replace c with x and note this on the scratch paper. The question states that the two expressions are equivalent, so enter the expressions with an equal sign between them.

Using the computer keyboard, type this into the entry field:

3 / Y + = X → = 1 5 % ? /
% 5 Y + = # 3) 0

The screen will show this:



Check out Section 2 of this guide to learn more about using the graphing calculator.

Notice that the graph display shows a vertical line through the value of 6, which means $x = 6$. For this question, $x = c$, so $c = 6$. The correct answer is (B).

7


Mark for Review

A total of 200 pets were adopted at an event. If 70% of the adopted pets were dogs, how many of the pets were dogs?

Here's How to Crack It

The question asks for a value based on a percentage. In this case, the percentage is 70% and the number of adopted pets is 200.

Using the computer keyboard, type this into the entry field:

&	)	%	@	)	)
7	0	5	2	0	0

Recall that the built-in calculator will automatically enter the word “of” after a percent sign. That helps a lot because *of* means multiplication, so the calculator is putting in the math operation for you. And then it solves it for you.



That’s all you have to do! The correct answer is 140.



15  Mark for Review

A postal machine processes mail at a constant rate of 21 pieces of mail per minute. At this rate, how many pieces of mail would the machine process in 7 minutes?

(A) 3

(B) 14

(C) 28

(D) 147

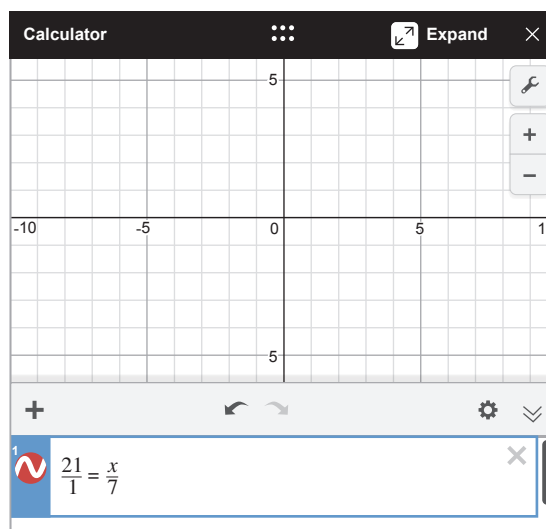
Here’s How to Crack It

The question asks for a value given a rate. Begin by reading the question to find information about the rate. The question states that the machine *processes mail at a constant rate of 21 pieces of mail per minute*. Set up a proportion to determine how many pieces of mail the machine will process in 7 minutes, being sure to match up units. The proportion is $\frac{21 \text{ pieces of mail}}{1 \text{ minute}} = \frac{x \text{ pieces of mail}}{7 \text{ minutes}}$. Now that you’ve done the work of reading the question carefully and setting up the math on your scratch paper, either solve it by hand or use the calculator. The calculator gives you a visual solution to this equation.

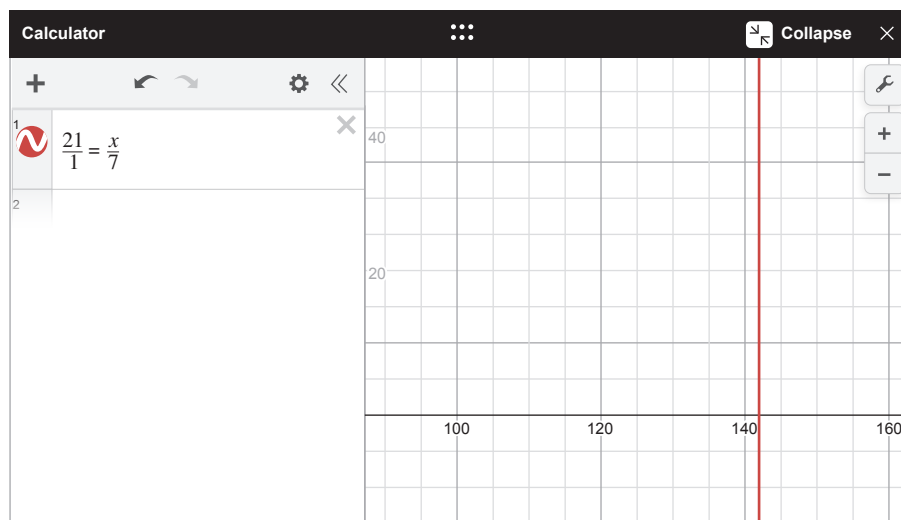
Using the computer keyboard, type this into the entry field:

@ 2 ! 1 ? / ! 1 → + = × ? / & 7

The screen will show this:



In this case, the visual solution does not show up in the default graph display, so expand the calculator and use the zoom buttons to zoom out until the result is visible:



The graph display shows a line between 140 and 160, and the only answer in that range is 147. The correct answer is (D).

You might decide that solving on the scratch paper is easier in the example above, and that's fine! One of your goals while you practice with the built-in calculator is to decide when it's going to be simpler and faster to perform the calculations by hand. Use the calculator when you know it will speed you up and improve your accuracy. Don't make careless mistakes by doing math in your head.

Now that you're comfortable with using the calculator for some relatively straightforward questions, let's look at a few that are longer or test more advanced topics.

This example is from the first module of Test 1.

7

Mark for Review

A random sample of 5,000 students out of 60,000 undergraduate students at a university were surveyed about a potential change to the registration system. According to the survey results, 75% of the respondents did not support the existing registration system, with a 4% margin of error. Which of the following represents a reasonable total number of students who did not support the existing registration system?

(A) 1,250

(B) 3,750

(C) 13,800

(D) 43,800

Here's How to Crack It

Before entering anything into the calculator, read the question carefully to make sure you know what to do. The question asks for a reasonable number based on survey results and a margin of error. Work in Bite-Sized Pieces and eliminate after each piece. Work with the survey results first. Start by applying the percent of respondents who did not support the existing registration system to the entire population of undergraduate students.

Enter the following into the calculator:

& % % ^))))
7 5 5 6 0 0 0 0

Recall that the built-in calculator will automatically enter the word “of” after a percent sign. You can also see that longer numbers have small spaces every 3 digits instead of commas.

1 75% of 60 000 ×

= 45 000

The result of the calculation is 45,000. Eliminate (A) and (B) because they are not close to this value and do not represent a reasonable number of students who did not support the existing registration system. Now deal with the margin of error, which expresses the amount of random sampling error in a survey’s results. The margin of error is 4%, meaning that results within a range of 4% above and 4% below the estimate are reasonable. A 4% margin of error will not change the result very much, and (D) is the only answer choice close to 45,000. To check, calculate the lower limit of the range based on the margin of error, since 43,800 is less than 45,000. To find the lower limit, subtract 4% from 75% to get 71%, then find 71% of the total population.

Using the computer keyboard, type into the entry field:

& ! % ^))))
7 1 5 6 0 0 0 0

The screen will show this:

1 71% of 60 000 ×

= 42 600

The result of the calculation is 42,600, so this number is the lower limit. The value in (C) is less than the lower limit, so it is not a reasonable number. Choice (D) contains a value between 42,600 and 45,000, so it is reasonable. The correct answer is (D).

The first example question in this section was a straightforward median question. Take a look at a question that takes another statistical concept, mean, and makes things more complicated.

This one is from the harder second module of Test 1.

20



Mark for Review

114, 109, 106, 111

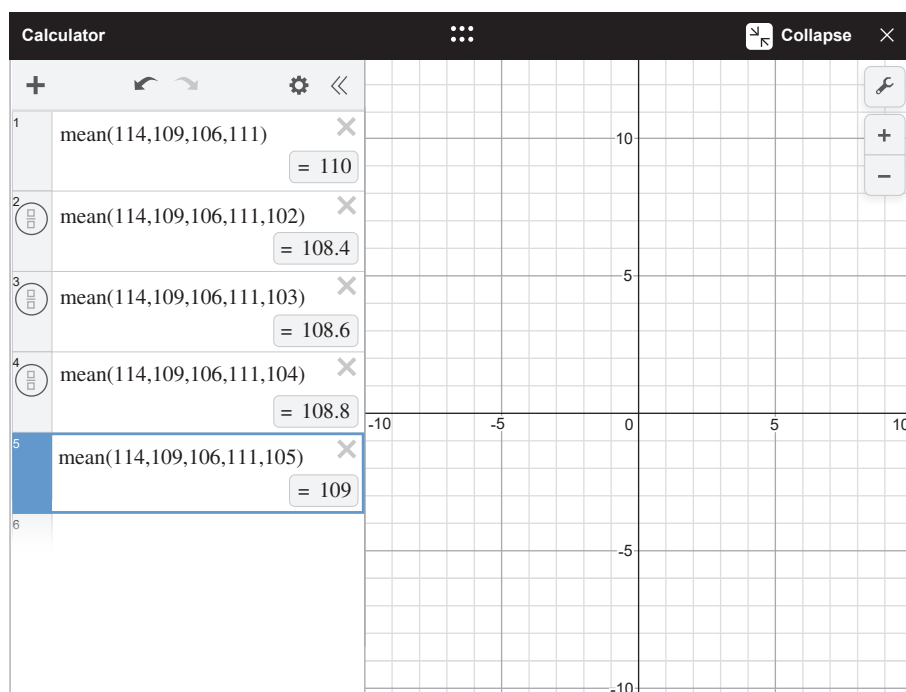
A data set consists of 5 positive integers greater than 101. What is the value of the smallest integer in the data set if the mean of the entire data set is an integer that is less than the mean of the four integers from the data set shown above?

Here's How to Crack It

The question asks for a value given information about the mean, or average, of a data set. Click on the **functions** button to open the menu, then scroll down a little to find the **mean** button. You can also type the word “mean” followed by parentheses. Enter the four numbers from the data set inside the parentheses to find the mean.



The question asks for the smallest integer greater than 101 that makes the mean of the data set an integer smaller than the mean of the first four numbers, which is 110. Starting with the smallest integer greater than 101, which is 102, find the mean of the new data set. Keep adding 1 to the new fifth integer until the mean is an integer less than 110.



Be sure to enter the value of the new fifth integer as your answer, not the value of the new mean. The correct answer is 105.

Can the calculator help with geometry questions? Sometimes, yes! When you see a geometry question, usually the best thing to do is to follow the Geometry Basic Approach: draw a figure, label the figure, and write down formulas. However, some questions that look like geometry are really about ratios or proportions. When you notice that, set things up carefully and let the calculator do some of the work for you.

The next example is from the harder second module of Test 1.

6



Mark for Review

A triangle has a base that is 65% of its height. If the base were decreased by 13 inches, how would the height need to change to keep the same proportions?

(A) It must increase by 13 inches.

(B) It must increase by 20 inches.

(C) It must decrease by 13 inches.

(D) It must decrease by 20 inches.

Here's How to Crack It

The question asks for the change in a value given a proportion. Since no specific numbers are given for the base and the height, plug in. Make the height of the larger triangle 100, so the base would be 65% of 100, or 65. Use the first entry field to set up a proportion. Put the base, 65, in the numerator and the height, 100, in the denominator. On the other side of the equal sign, enter the new base in the numerator and the unknown new height in the denominator. Since the base is decreased by 13, enter the new base as $(65 - 13)$. Use x or the variable of your choice to represent the new height. With the first proportion in the first entry field, use the second entry field to cross-multiply and solve.

1 $\frac{65}{100} = \frac{(65 - 13)}{x}$

2 $\frac{100(65 - 13)}{65} = 80$

Since the original height was 100, the change is $100 - 80 = 20$. The new height is less than the original height, so it decreased by 20. The correct answer is (D).

You might have noticed the term “plug in” in the previous example. If you’re not sure what that is, it’s time to work through the Other Digital SAT Algebra Strategies chapter in your *SAT Prep* book. Once you’re familiar with Plugging In and PITA, continue on to the last two examples.

The next example is from the first module of Test 1.

13



Mark for Review

Time (seconds)	Number of colonies of yeast
0	5
1	20
2	80
3	320

The table above shows the exponential growth of a type of yeast over time s , in seconds. There are c total yeast colonies on the count plate. What is the equation that represents this relationship, assuming that no yeast was added or removed after counting began?

(A) $c = (1 + 3)^s$

(B) $c = (1 + 5)^s$

(C) $c = 3(1 + 5)^s$

(D) $c = 5(1 + 3)^s$

Here’s How to Crack It

The question asks for the equation that represents the relationship between two variables. When given a table of values and asked for the correct equation, plug values from the table into the answer choices to see which one works. According to the table, $s = 2$ when $c = 80$. Enter the expression from each answer choice into the calculator, and replace s with 2. Eliminate any answers that do not equal 80. Choice (A) equals 16 when $s = 2$, so eliminate it.

1

$(1 + 3)^2$

×

= 16

Do the same thing with the remaining answers:

1	$(1 + 3)^2$	$\times$
		= 16
2	$(1 + 5)^2$	$\times$
		= 36
3	$3(1 + 5)^2$	$\times$
		= 108
4	$5(1 + 3)^2$	$\times$
		= 80

The expression in (D) equals 80 when $s = 2$, so the correct answer is (D).

The calculator didn't change your approach to the previous question. It just gave you an alternative to the scratch paper. When you use Plugging In on a question, use the first entry field to work out the target value, then use the next four fields to plug in the same value(s) and see which one matches the target. The calculator does the math, but it's still your responsibility to RTFQ and set things up correctly.

Try one more like that, this time from the harder second module of Test 1.

21 Mark for Review

A teacher awards points to a class based on completed assignments. He gives 5 points per assignment for the first 50 completed assignments and 3 points for each additional completed assignment beyond 50. When $a \geq 50$, which function g gives the total amount of points earned by the class for a completed assignments?

(A) $g(a) = 3a + 5$

(B) $g(a) = 3a + 100$

(C) $g(a) = 3a + 250$

(D) $g(a) = 8a - 150$

Here's How to Crack It

The question asks for the function that represents a certain situation. There are variables in the answer choices, and the question asks about the relationship between the number of points and the number of assignments, so plug in. Plug in a value for a that is greater than 50, such as $a = 52$. Use the first entry field to work the question in Bite-Sized Pieces: multiply the first 50 questions by 5, and multiply the 2 additional questions by 3. The target value is 256. Then use the next four entry fields to test each answer choice when $a = 52$.

1	$5 \cdot 50 + 3 \cdot 2$	$\times$
		= 256
2	$3 \cdot 52 + 5$	$\times$
		= 161
3	$3 \cdot 52 + 100$	$\times$
		= 256
4	$3 \cdot 52 + 250$	$\times$
		= 480
5	$8 \cdot 52 - 150$	$\times$
		= 266

Only (B) matches the target value of 256, so the correct answer is (B).

Conclusion

Those were just a few examples of questions that the built-in calculator can help you answer correctly and efficiently. Try using it on practice questions and practice tests to decide when and how you will use the calculator on test day. Section 4 contains example questions that utilize the built-in graphing calculator; make sure you're familiar with the information in Section 2 before moving on.

Section 4

Graphing Calculator Example Questions

The best way to practice with the graphing calculator is to use it for some Digital SAT questions. The examples in this chapter all come from Test 1 in your *Digital SAT Prep* book. If you haven't taken that test yet, you should do so before exploring these examples.

Let's start with a question that might not look like it's about graphing at all. This example is from the first module of Test 1.

8



Mark for Review

What is the negative solution to the equation

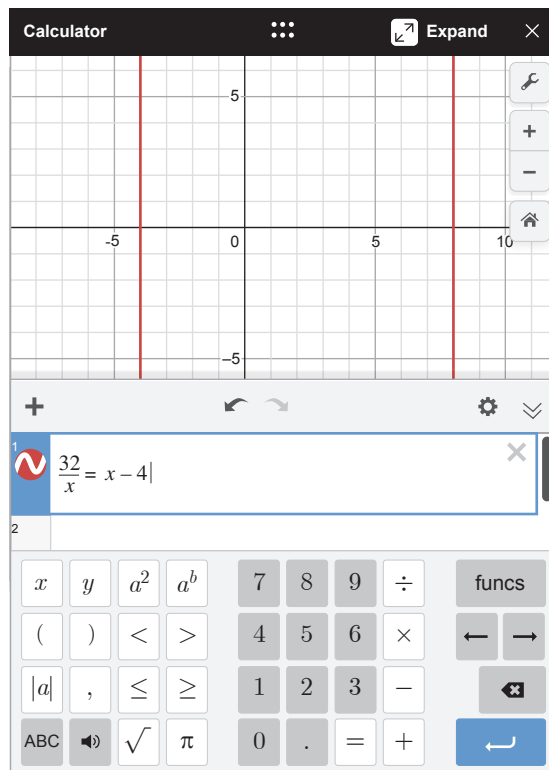
$$\frac{32}{a} = a - 4?$$

Here's How to Crack It

The question asks for a solution to an equation. To solve this algebraically, you would need to rearrange the equation into the standard form of a quadratic and then factor the quadratic. The graphing calculator doesn't require as much effort. First, change a to x because the calculator doesn't show a graph when you use variables other than x and y . Write $a = x$ on your scratch paper to keep track of this. Next, enter the equation as written:

#	@	?	x	→	+	x	−	\$
3	2	/			=		-	4

The screen will show this:



The graphing area shows two straight, vertical lines; those are the solutions to the equation. Zoom in to see that the lines cross the x -axis at -4 and 8 . The question asks for the negative solution, so the correct answer is -4 .



You saw in Section 2 of this guide how the graphing calculator can help you find the vertex and intercepts of a parabola. Here's a typical Digital SAT question that asks you to do just that. This example is also from the first module of Test 1.

10

Mark for Review

What is the x -intercept of the function $f(x) = (22)^x - 1$ when it is graphed in the xy -plane, where $y = f(x)$?

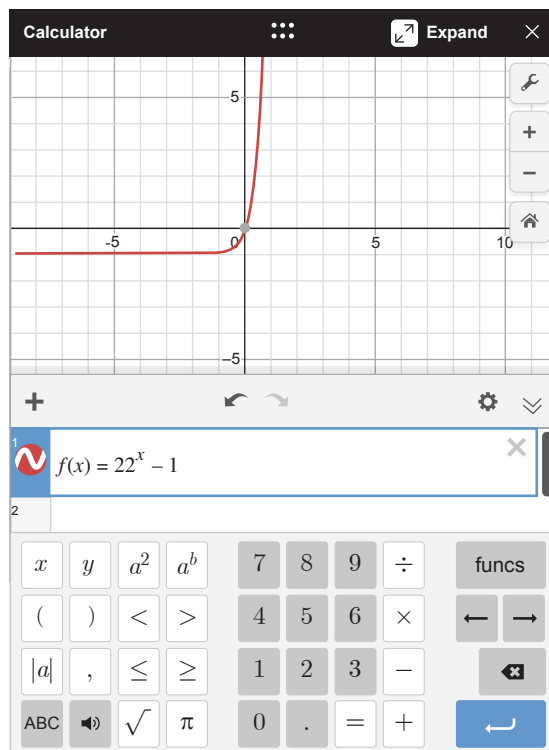
☐ (A) $(-1, 0)$
☐ (B) $(0, 0)$
☐ (C) $(21, 0)$
☐ (D) $(22, 0)$

Here's How to Crack It

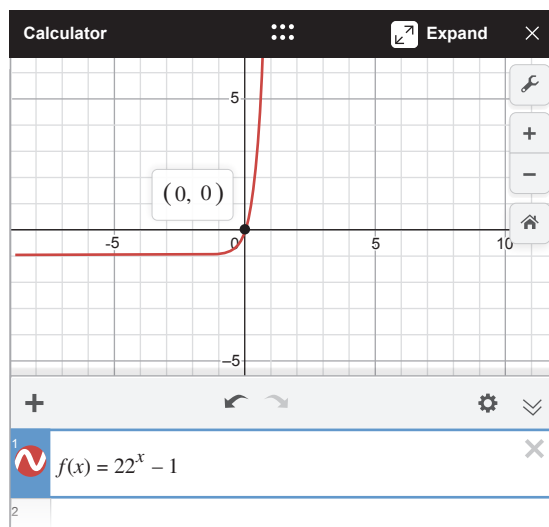
The question asks for the x -intercept of a function. Enter the function into the first entry field, being careful to use the right arrow key or the mouse after entering the exponent.

F (9 x) 0 + = (@ @) ^ x → - !

The screen will show this:



Click on the gray dot to see that the coordinates of the point where the graph crosses the x -axis are (0, 0).



The x -intercept is at (0, 0). The correct answer is (B).

Let's look at another question, this time about the maximum value of a function. Here's another example from the first module of Test 1.

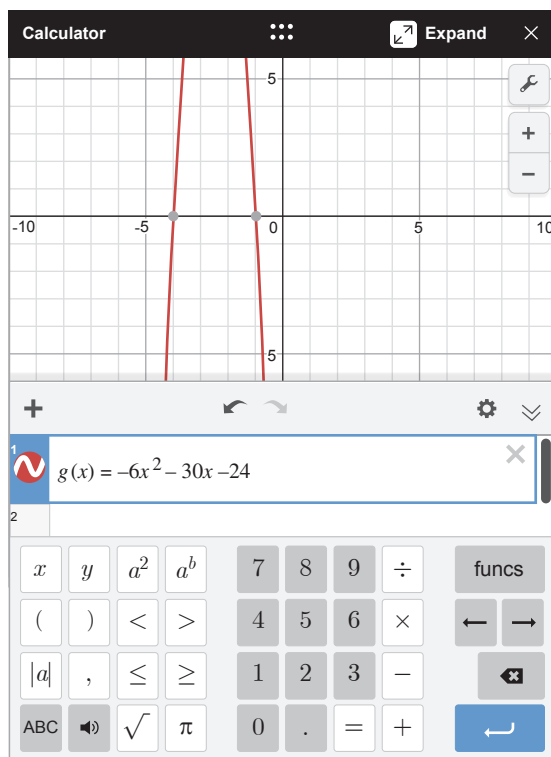
20

Mark for Review

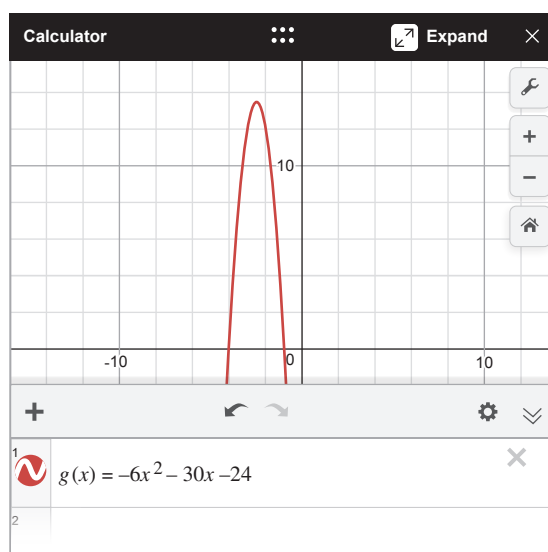
Function g reaches its maximum value when $x = a$. If $g(x) = -6x^2 - 30x - 24$, what is the value of a ?

Here's How to Crack It

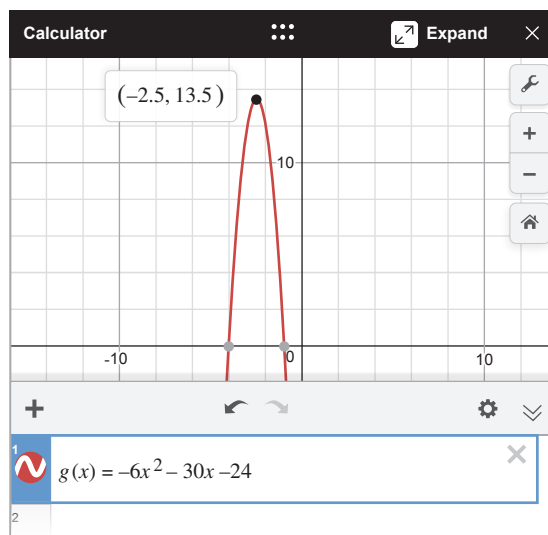
The question asks for the value when a quadratic function reaches its maximum. A parabola reaches its minimum or maximum value at its vertex, so find the x -coordinate of the vertex. Enter the function into the graphing calculator, and the screen will show this:



The vertex isn't visible, so zoom out or click and drag the window until you can see it.



Either click in the entry field with the equation or on the graph to show the gray dot at the vertex, then click the dot to show the coordinates of that point.



The question asked for the value of a , which is the x -coordinate of the vertex. The correct answer is -2.5 .

One of the great things about graphing these equations with the built-in calculator is that finding multiple points doesn't require much more work than finding a single point: once you've put the equation into the calculator, most of the work is already done.

Take a look at a question from the harder second module of Test 1. This is question 16 out of 22, which means the test-writers think this is a pretty hard question. However, the graphing calculator saves you a lot of time and energy.

16



Mark for Review

$$3x - 4y = 17$$

In the xy -plane, the graph of a line with an x -intercept of $(c, 0)$ and a y -intercept of $(0, k)$, where c and k are constants, can be represented by the equation above. What is the value of $\frac{c}{k}$?

(A) $-\frac{4}{3}$

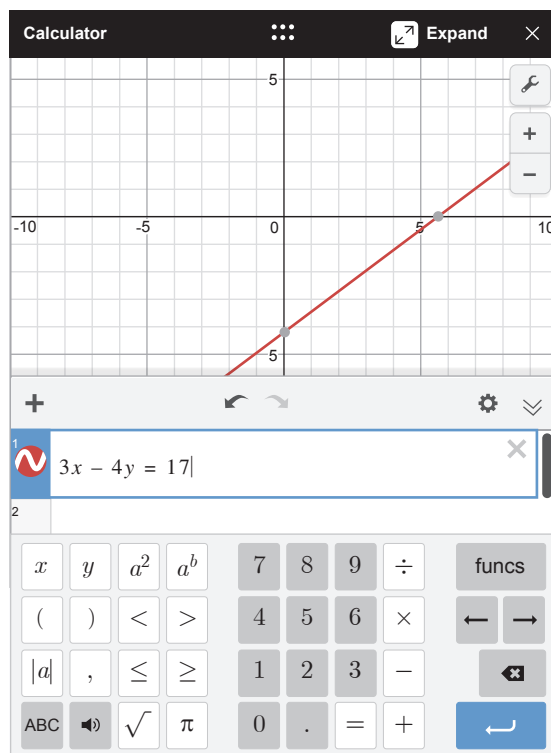
(B) $-\frac{3}{4}$

(C) $\frac{3}{4}$

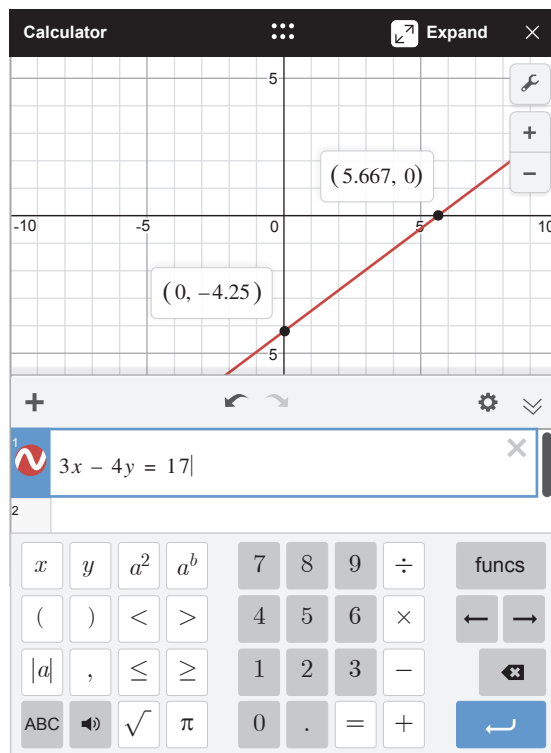
(D) $\frac{4}{3}$

Here's How to Crack It

The question asks for the value of an expression given the equation of a graph in the xy -plane. The x -coordinate of the x -intercept is c , and the y -coordinate of the y -intercept is k , so use the graphing calculator to find those two values. Enter the equation into the graphing calculator.

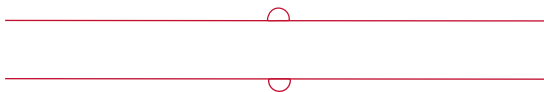


Click on the gray dots to find the exact values of the x - and y -intercepts.



Plug in 5.667 for c and -4.25 for k , and use the calculator to solve:

$$\frac{c}{k} = \frac{5.667}{-4.25} \approx -1.33 \text{ or } -\frac{4}{3}. \text{ The correct answer is (A).}$$



The next question is also from the harder second module of Test 1. Let's see if the graphing calculator makes this one easier than it looks, too.

11



Mark for Review

$$3x^2 - y - 26 = 0$$

$$y = -3x + 10$$

The point (a, b) is an intersection of the system of equations above when graphed in the xy -plane.

What is a possible value of a ?

(A) -4

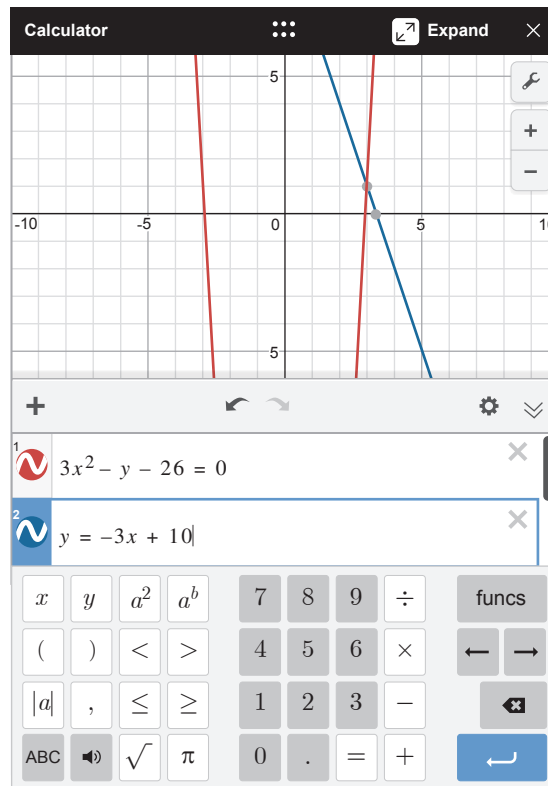
(B) 6

(C) 20

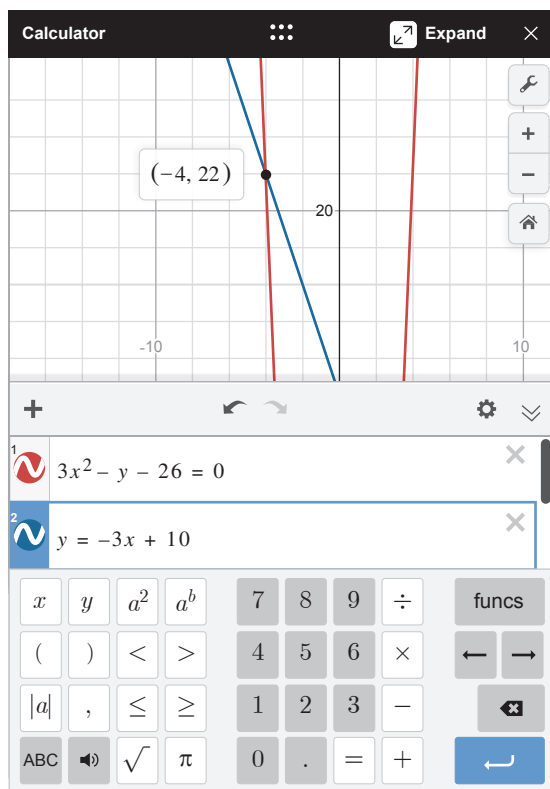
(D) 26

Here's How to Crack It

The question asks for the value of the x -coordinate of the solution to a system of equations. To find the intersection, simply enter the equations into the calculator.



One of the points of intersection is visible at $(3, 1)$. The question asks for the x value at the intersection, so $x = a = 3$, but 3 isn't an answer choice. Zoom out or click-and-drag the window to see the other intercept.



Clicking on the dot at the intercept shows that the coordinates are $(-4, 22)$. Now $x = a = -4$. The correct answer is (A).

Finally, graphing can be incredibly powerful when combined with Plugging In the Answers (PITA) or Plugging In. If you're not sure what those things are, work through the Other Digital SAT Algebra Strategies chapter in your *Digital SAT Prep* book, and then come back to this last example.

Put all your skills together on this question from the harder second module of Test 1.

15

Mark for Review

If c is a constant in the equation $10x^2 + c = -5x$, and the equation has no real solutions, what is the value of c ?

(A) -20

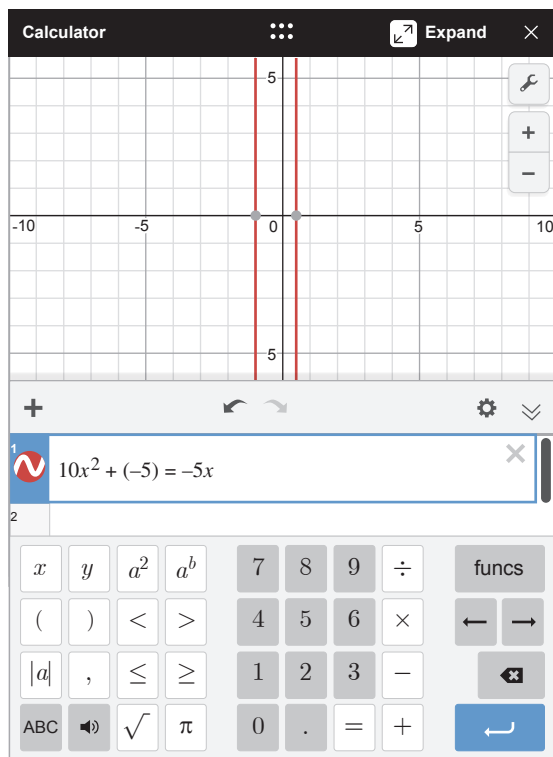
(B) -5

(C) 0

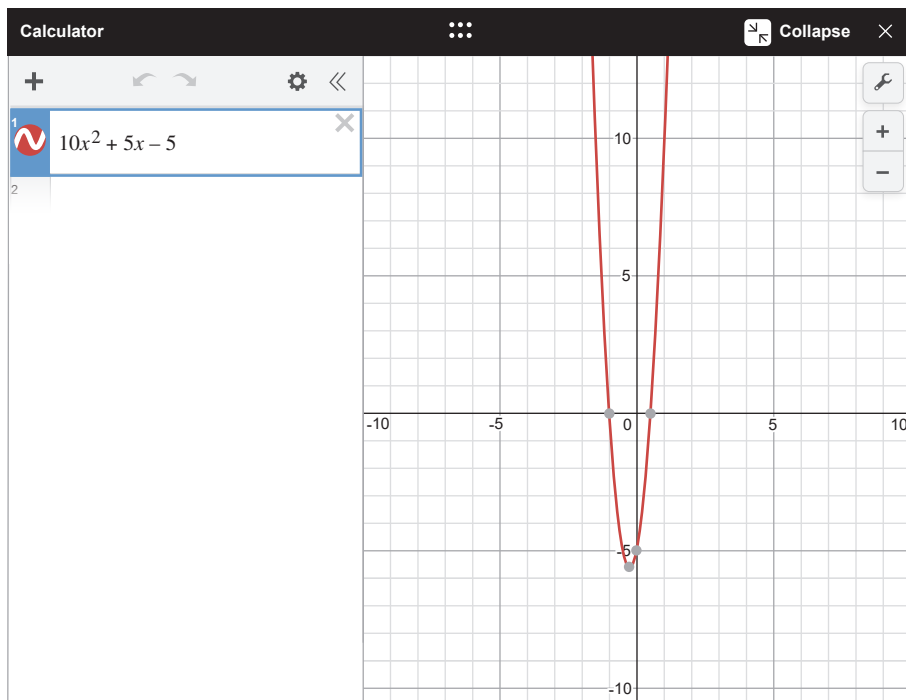
(D) 1

Here's How to Crack It

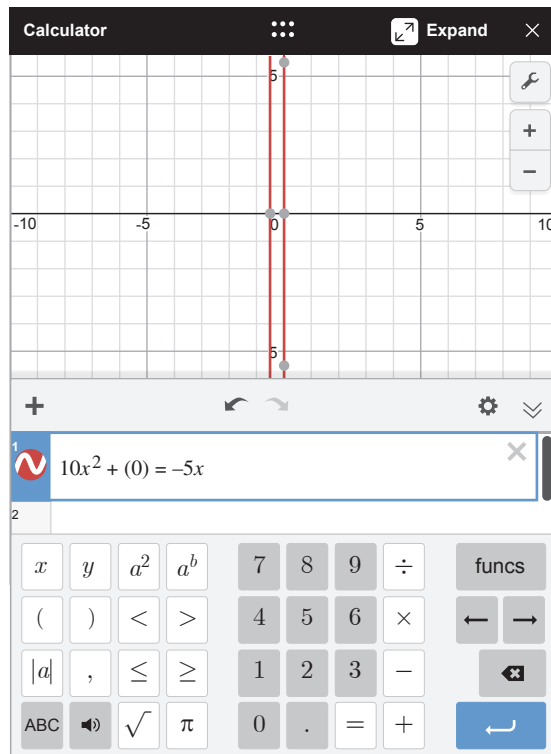
The question asks for the value of a constant in a quadratic equation. Any question that asks for the value of a constant that causes certain conditions to occur—such as the equation having no real solutions—is a great chance to use PITA. A graph of an equation with no real solutions will not intersect the x -axis. Plug in the answers for c and see what happens to the graph. Start with (B), and plug in $c = -5$. The equation becomes $10x^2 + (-5) = -5x$; enter this into the first entry field.



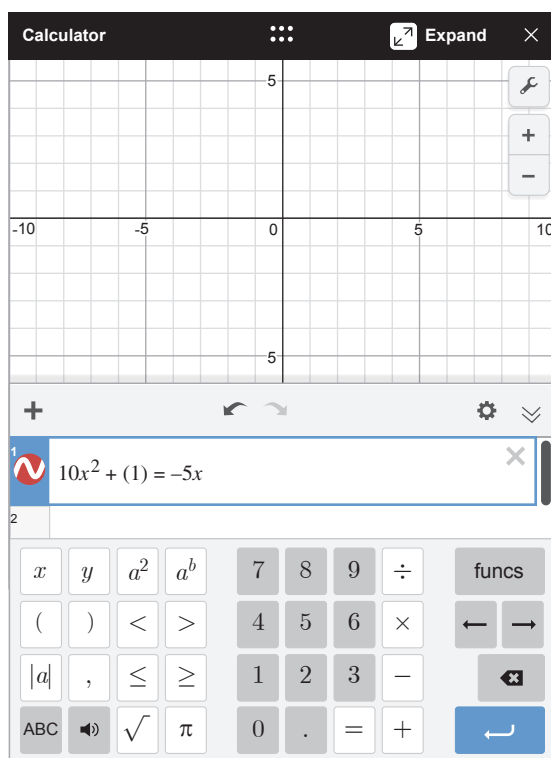
If you prefer to see the graph of a parabola, convert the equation into standard form by adding $5x$ to both sides of the equation. The equation becomes $10x^2 + 5x - 5 = 0$. Enter the expression on the left side of the equation, leaving out " $= 0$," to see this graph:



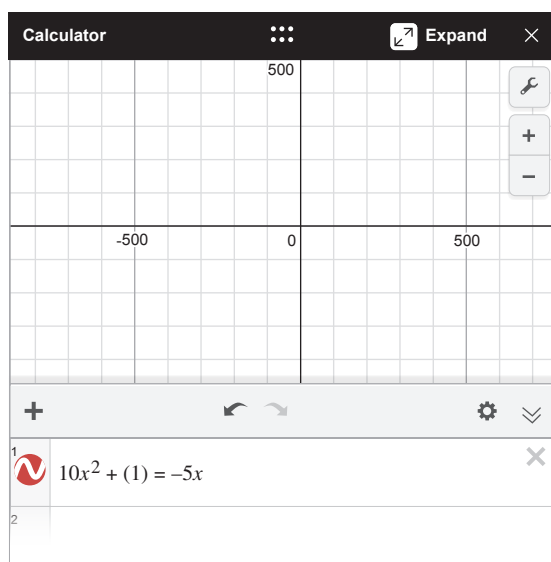
In both forms, the graph clearly intersects the x -axis twice, so it has 2 solutions. The correct answer should have no solutions, so eliminate (B). Try (C), and plug in $c = 0$. To save time, delete -5 from what's already in the entry field for c and replace it with 0.



The graph still intersects the x -axis twice, so eliminate (C). Try (D) and replace 0 with 1 for c .

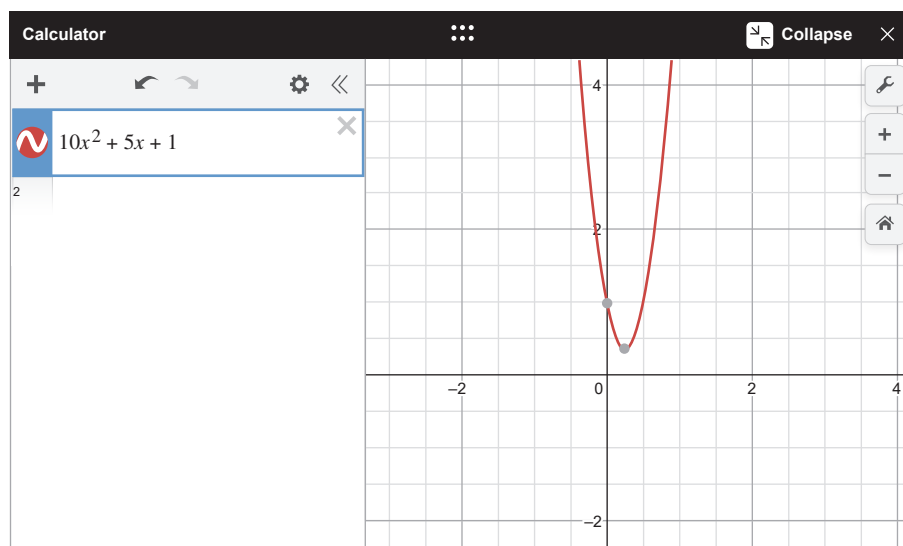


The graph seems to have disappeared. Zoom out and click and drag the graph to confirm.



There is no graph because the calculator cannot determine any real values that satisfy the equation. The equation has no real solutions.

If you have been using the standard form of a quadratic without “= 0” to graph a parabola, you will see a graph when $c = 1$.



However, the graph does not intersect the x -axis, so there are no solutions.

Whichever way you graph it, there are no solutions when $c = 1$. The correct answer is (D).



Conclusion

As you've seen, the built-in graphing calculator can help you avoid long, messy calculations and algebra, which means fewer mistakes. It also provides a visual to make things easier to see and interpret. Keep practicing with the calculator throughout the remainder of your prep, and feel free to come back to this guide at any time for a refresher.